Diseño y Desarrollo de servicios web – proyecto

John Alejandro Soto Gómez

Mi programa de formación

Tecnología en Análisis y Desarrollo de Software – ADSO

GA7-220501096-AA5-EV03

Instructor

Gustavo Adolfo Chacón

Servicio Nacional de Aprendizaje (SENA)

20 febrero de 2026

Contenido

Introducción……..…….…….…….…..… 3
Objetivos…….……..……..……..…….…..… 3
Alcance……..……. 3
Desarrollo de la actividad ……..…….…….…… 4
Conclusiones……..…….28
Referencias Bibliográficas……..……..29

Introducción

El presente proyecto tiene como finalidad desarrollar e integrar una aplicación web para la gestión de usuarios dentro del sistema Xenmeteo, utilizando una arquitectura cliente-servidor. La solución implementa un frontend desarrollado en HTML y JavaScript, el cual se comunica mediante peticiones HTTP con un backend construido con Spring Boot. Este backend se encarga de procesar las solicitudes, aplicar la lógica de negocio y persistir la información en una base de datos relacional gestionada por MySQL.

La integración entre estas tecnologías permite evidenciar el funcionamiento completo del flujo de datos, desde la interacción del usuario en el navegador hasta el almacenamiento definitivo en la base de datos, cumpliendo con los principios de arquitectura en capas y separación de responsabilidades.

Objetivos

Desarrollar e integrar un sistema de registro de usuarios para la plataforma Xenmeteo, mediante la implementación de un frontend en HTML y JavaScript conectado a un backend en Spring Boot, garantizando el almacenamiento correcto de la información en una base de datos MySQL.

Alcance

El proyecto contempla el desarrollo e integración del módulo de registro de usuarios para la plataforma Xenmeteo bajo una arquitectura cliente-servidor. Incluye la creación de una interfaz web en HTML y JavaScript que permite capturar los datos del usuario y enviarlos al servidor mediante peticiones HTTP tipo POST.

En el backend se implementa un servicio REST utilizando Spring Boot, encargado de recibir, procesar y gestionar la información. La persistencia de datos se realiza mediante JPA e Hibernate, almacenando la información en una base de datos relacional administrada por MySQL.

El alcance incluye la configuración de la conexión a la base de datos, la creación de la entidad Usuario, el repositorio de acceso a datos y la validación del flujo completo entre frontend, backend y base de datos. No contempla mecanismos avanzados de seguridad, autenticación, cifrado de contraseñas ni despliegue en entornos productivos, ya que el objetivo principal es demostrar la correcta integración y almacenamiento funcional de la información.

Desarrollo de la actividad

Diseño y Desarrollo de servicios web – proyecto

Contextualización dentro de Xenmeteo

Xenmeteo es una plataforma web que permite:

- Consultar pronóstico meteorológico
- Visualizar calidad del aire en tiempo real
- Analizar series temporales
- Gestionar usuarios

Objetivo

Desarrollar un servicio web (API REST) que permita:

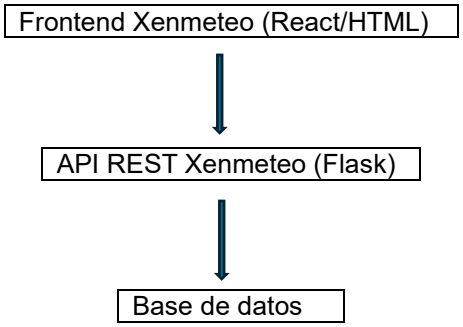
1. Registro de usuario
2. Inicio de sesión
3. Validación de autenticación
4. Manejo de errores

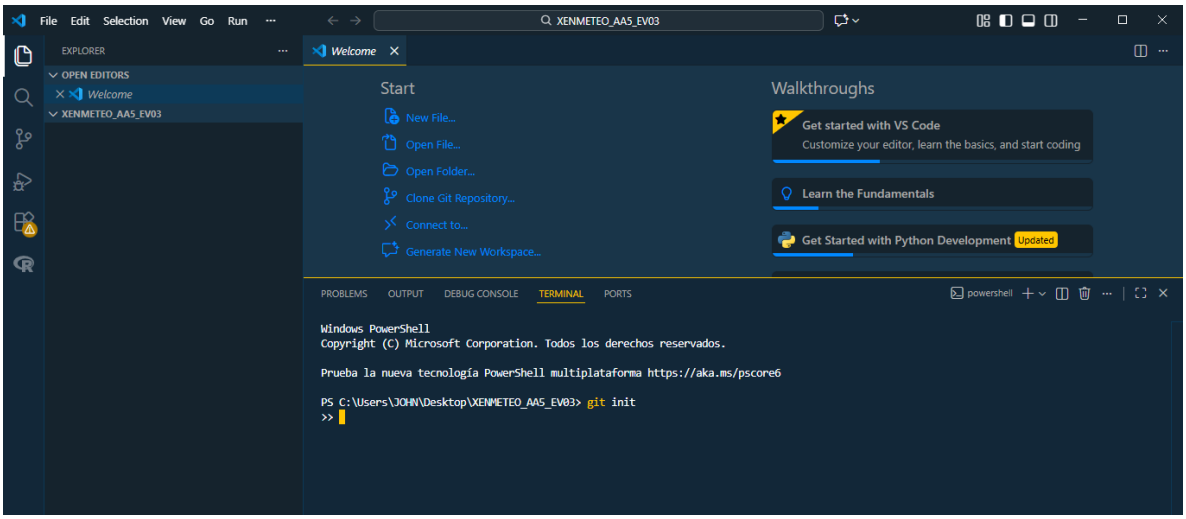
- **Datos de entrada:**
- usuario
- contraseña

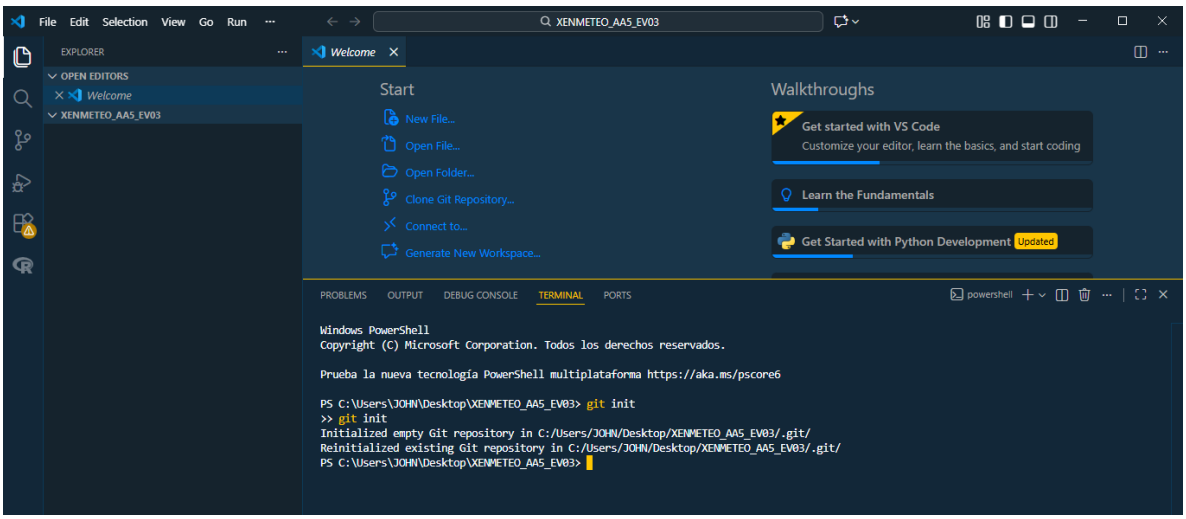
- **Respuesta esperada:**
- Si autenticación correcta → mensaje: "Autenticación satisfactoria"
- Si falla → "Error en la autenticación"

Diseño de la API dentro de Xenmeteo

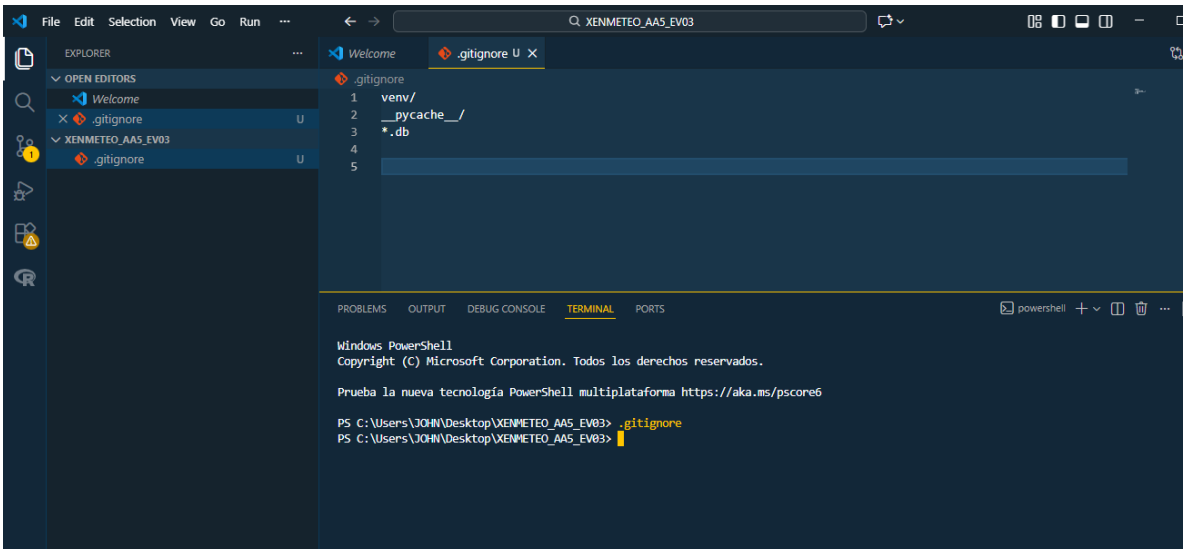
Arquitectura



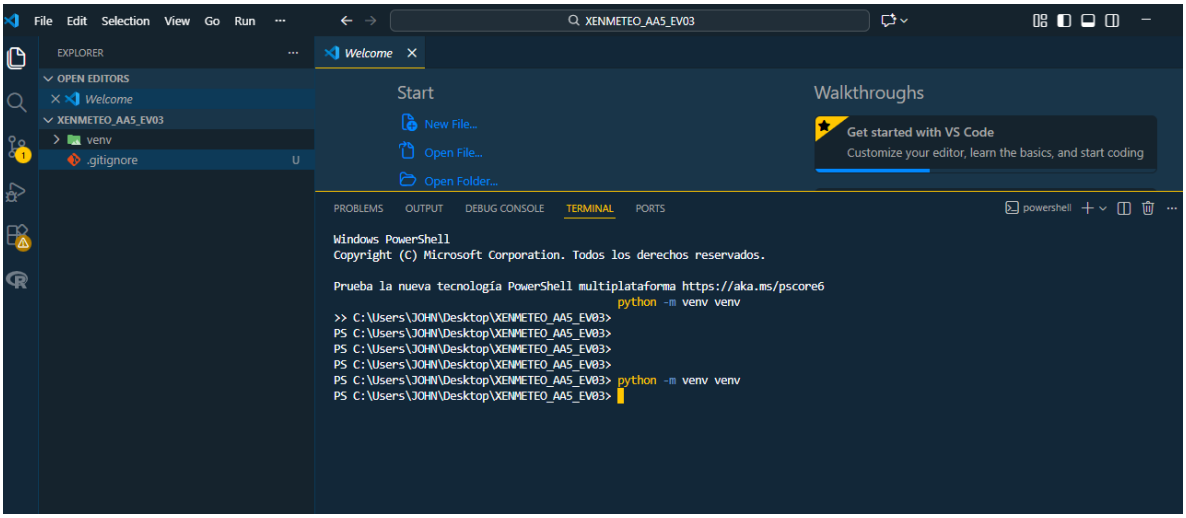




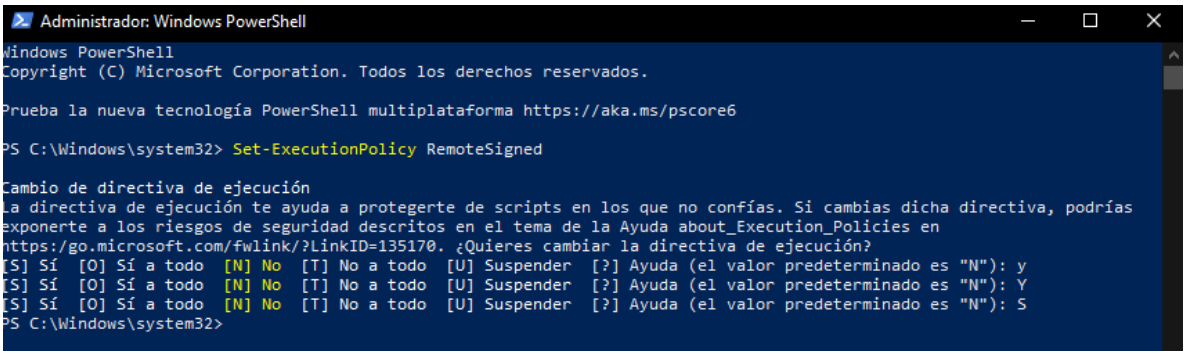
Creo un archivo .gitignore

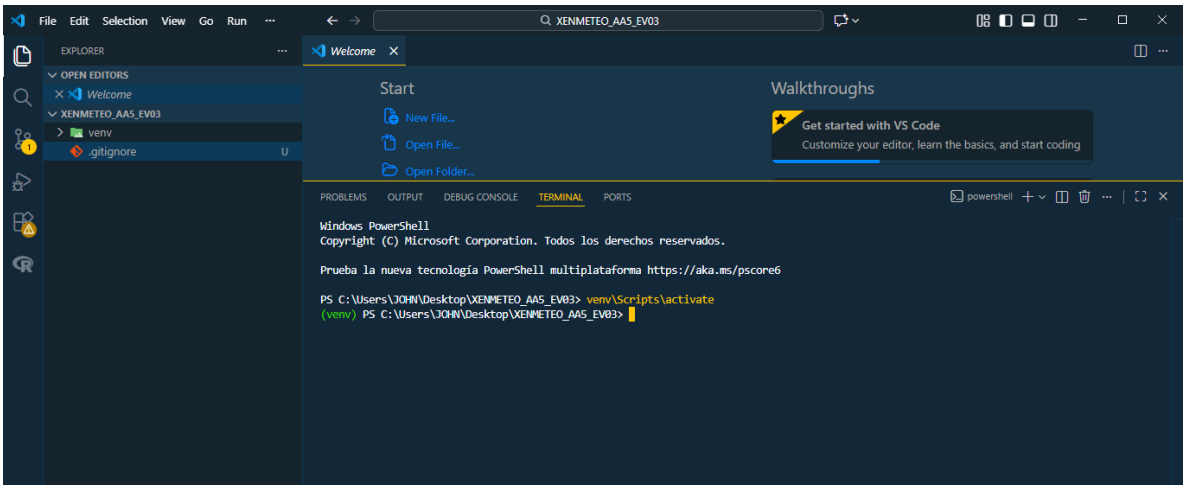


Crear entorno virtual



Activamos entorno virtual en powershell



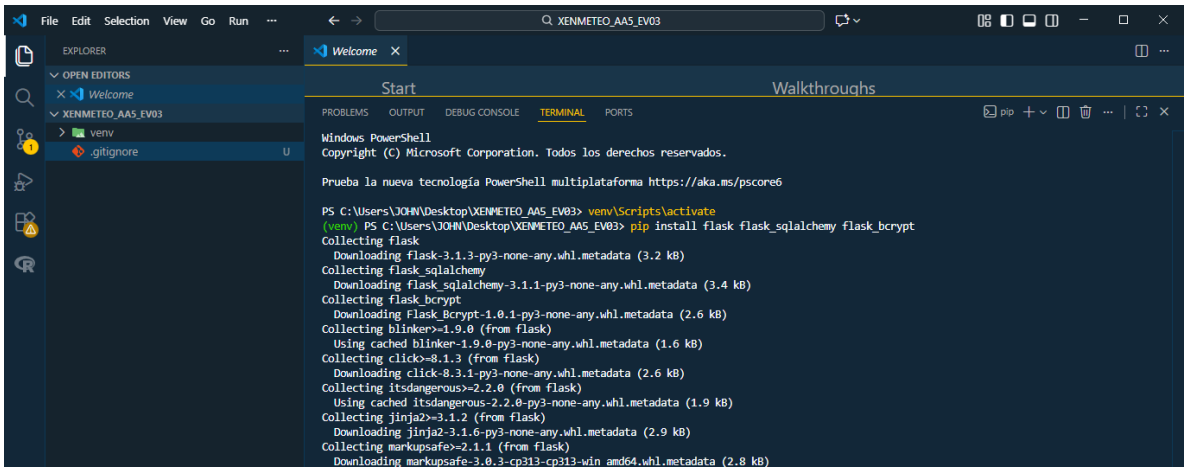


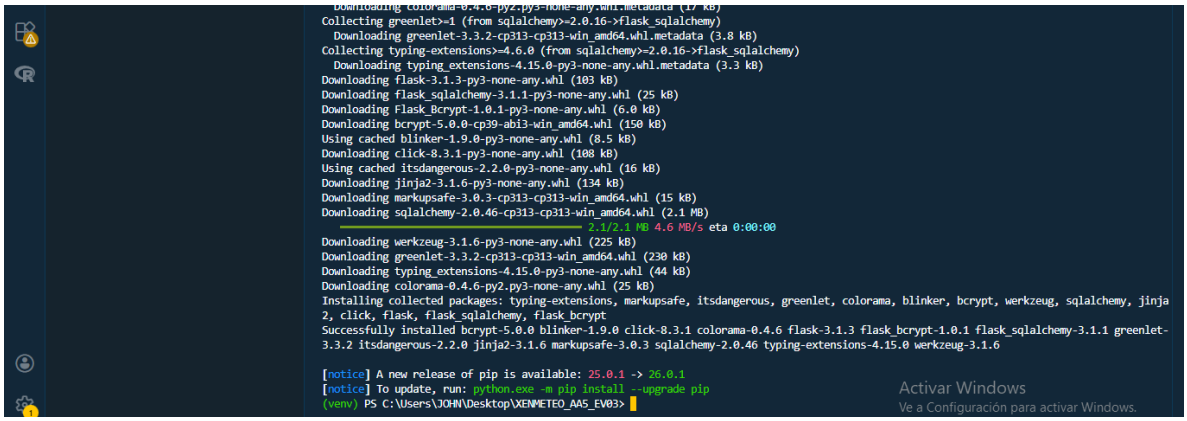
Con lo anterior, Windows por seguridad bloquea archivos .ps1 (PowerShell scripts).

El entorno virtual usa uno de esos scripts para activarse.
Al cambiar a RemoteSigned, le digo:
Permitir scripts locales creados en este equipo. Es totalmente seguro para desarrollo.

Instalar librerías necesarias

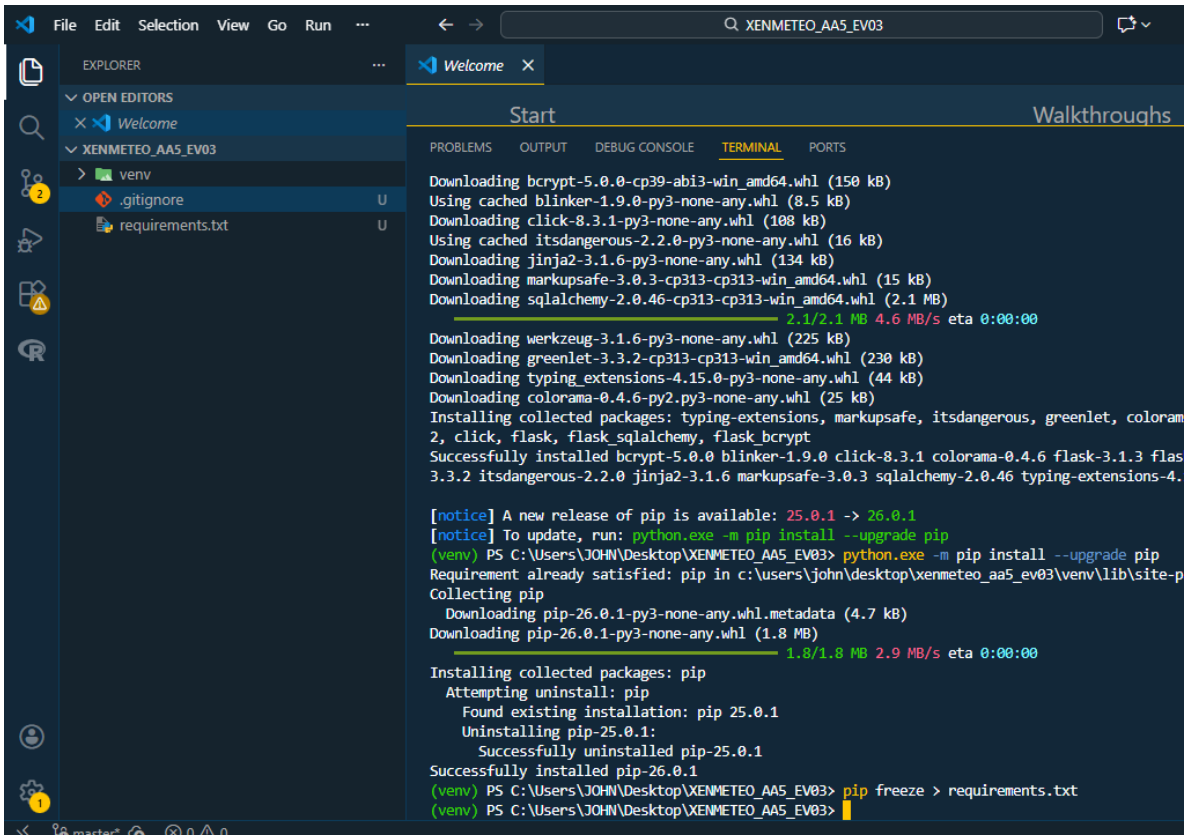
- Instalar Flask y base de datos



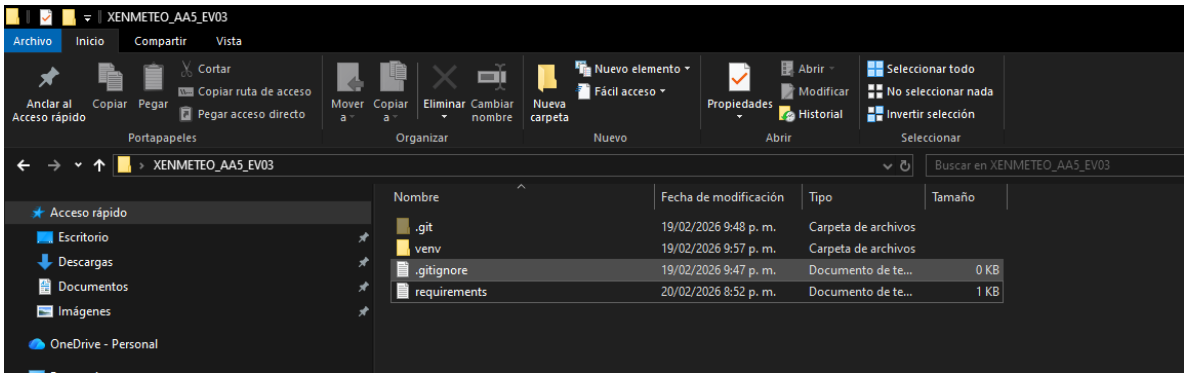


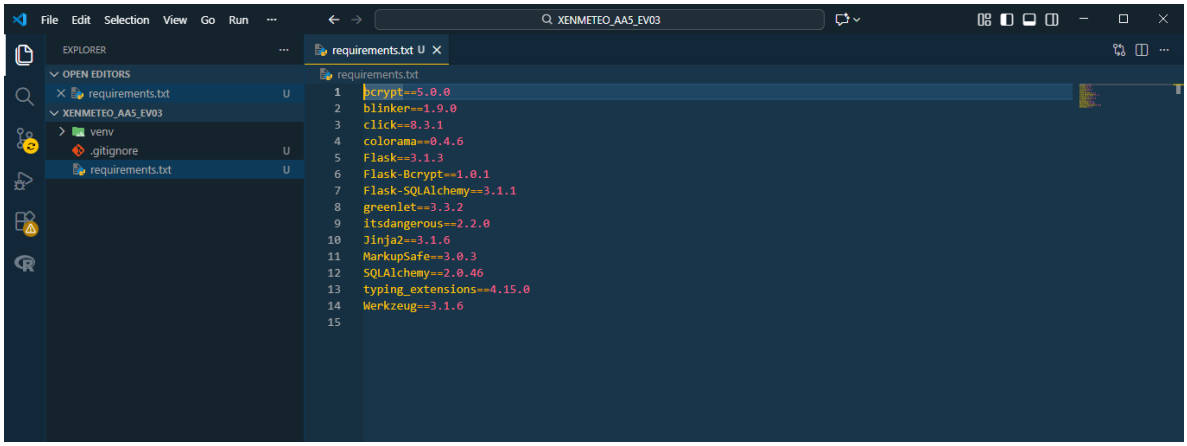
Luego, ponemos “python.exe -m pip install --upgrade pip” para actualizar cambios

Guardamos dependencias



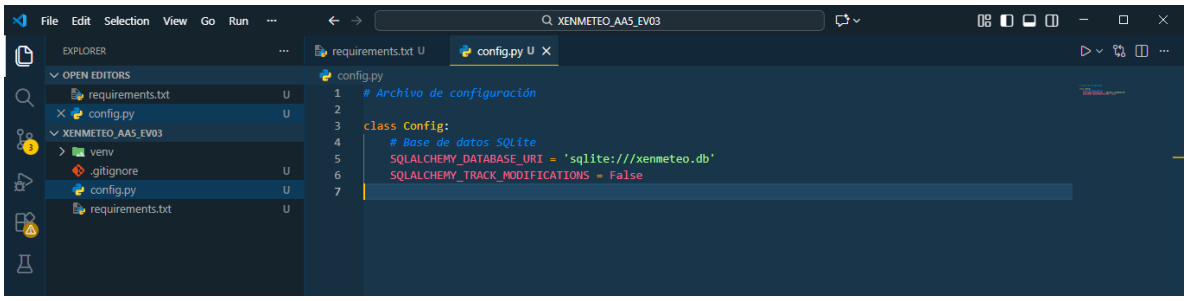
Ahora, ya se tiene el archivo requirements.txt



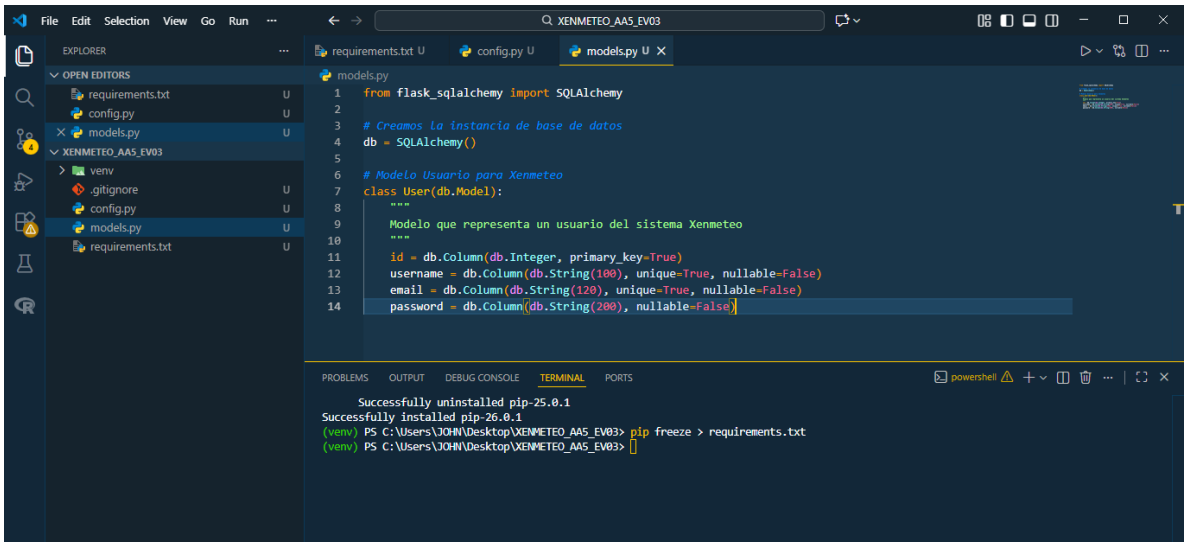


Creamos archivos del proyecto

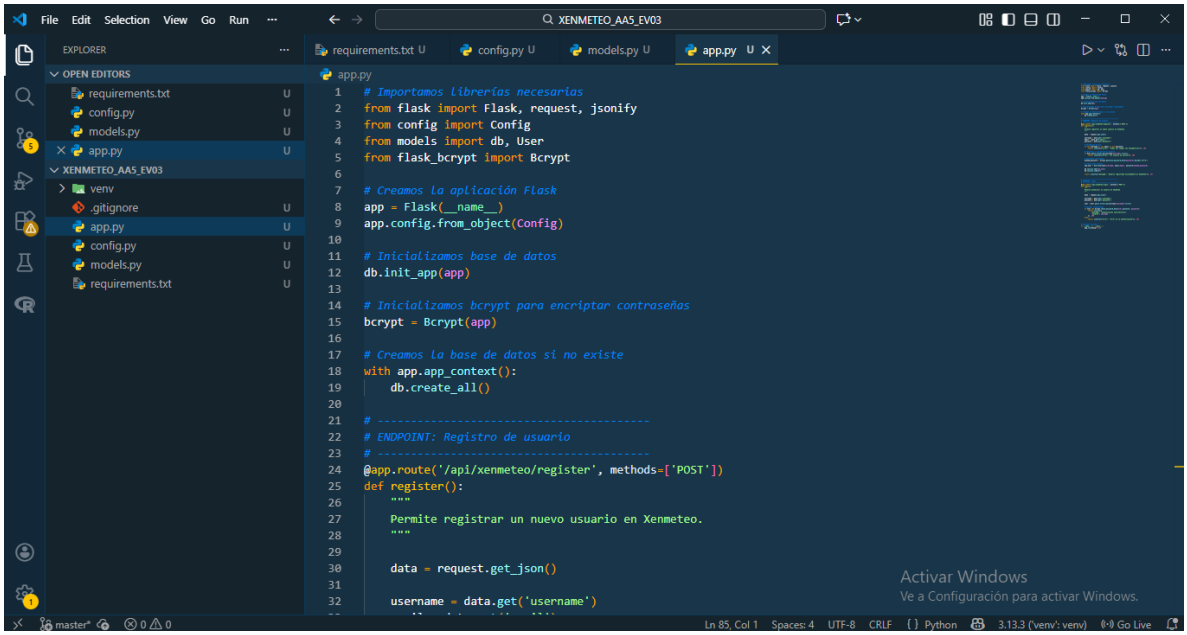
Creamos archivo config.py y copiamos lo siguiente:



Creamos archivo models.py y copiamos esto:

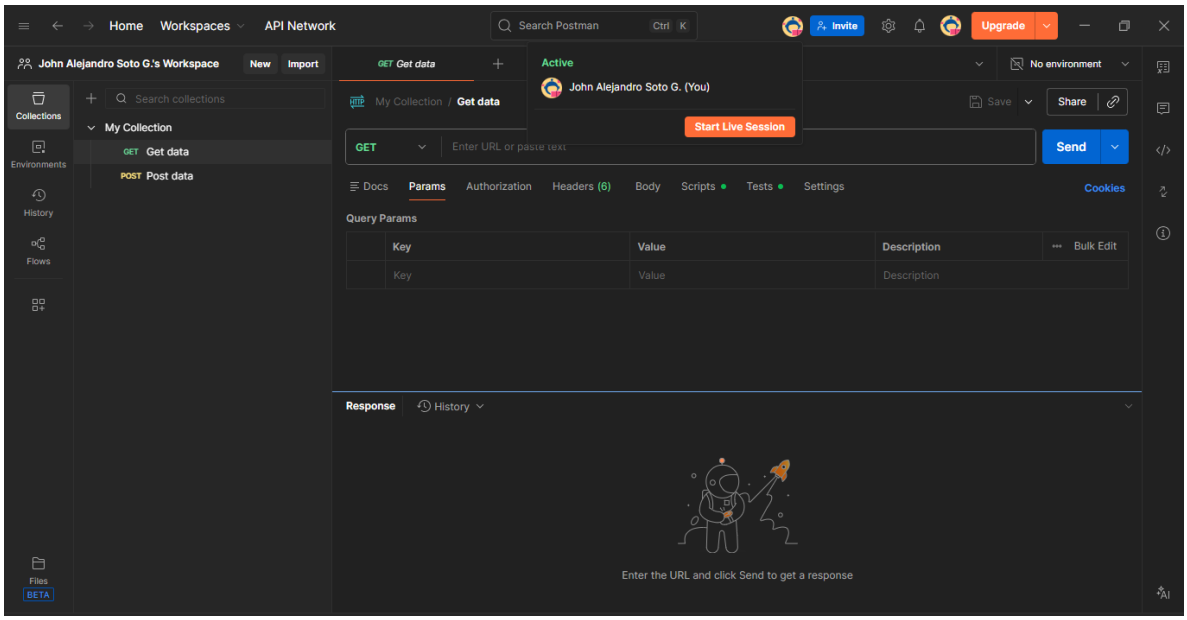


Creamos el archivo principal app.py

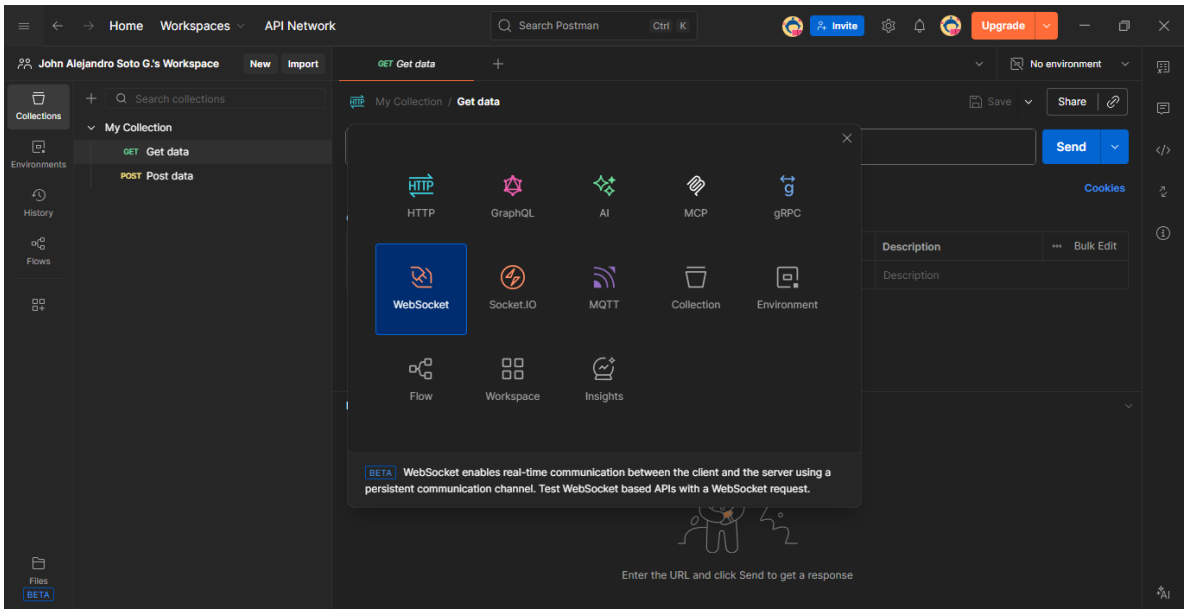


Comprobemos URL en Postman dado que es mi API en modo endpoint

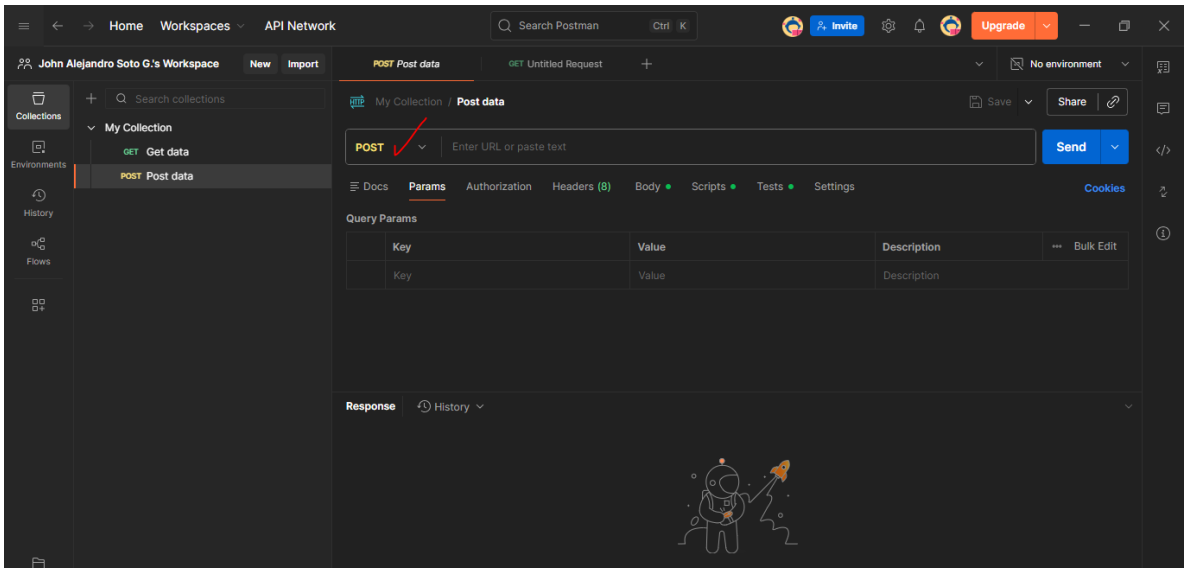
Abrimos Postman



New → HTTP Request



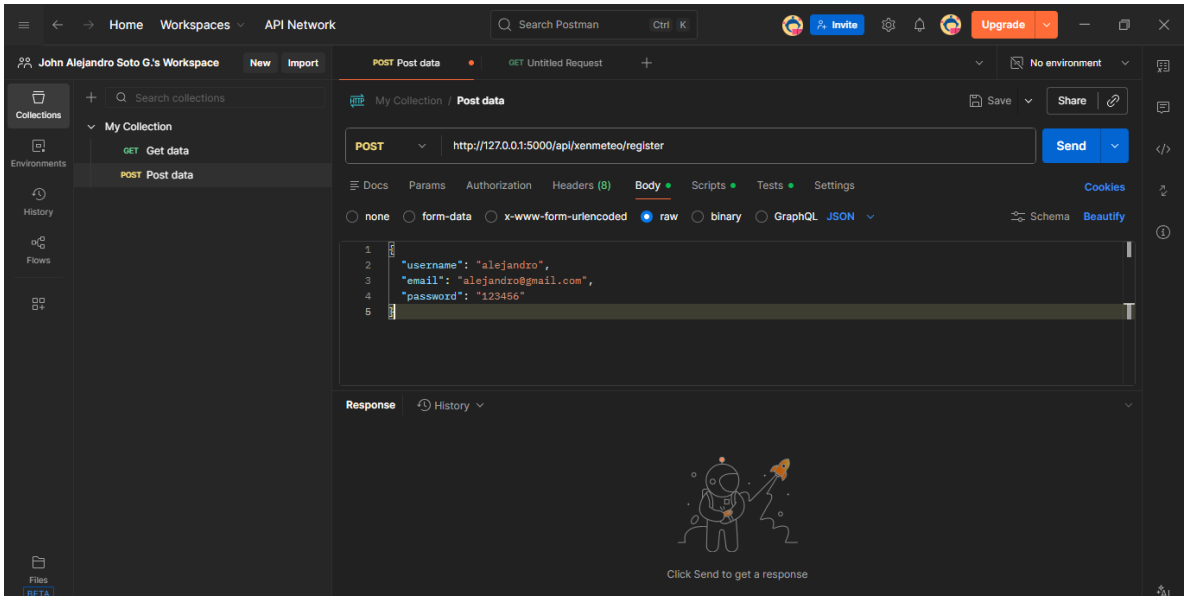
Configuramos el método



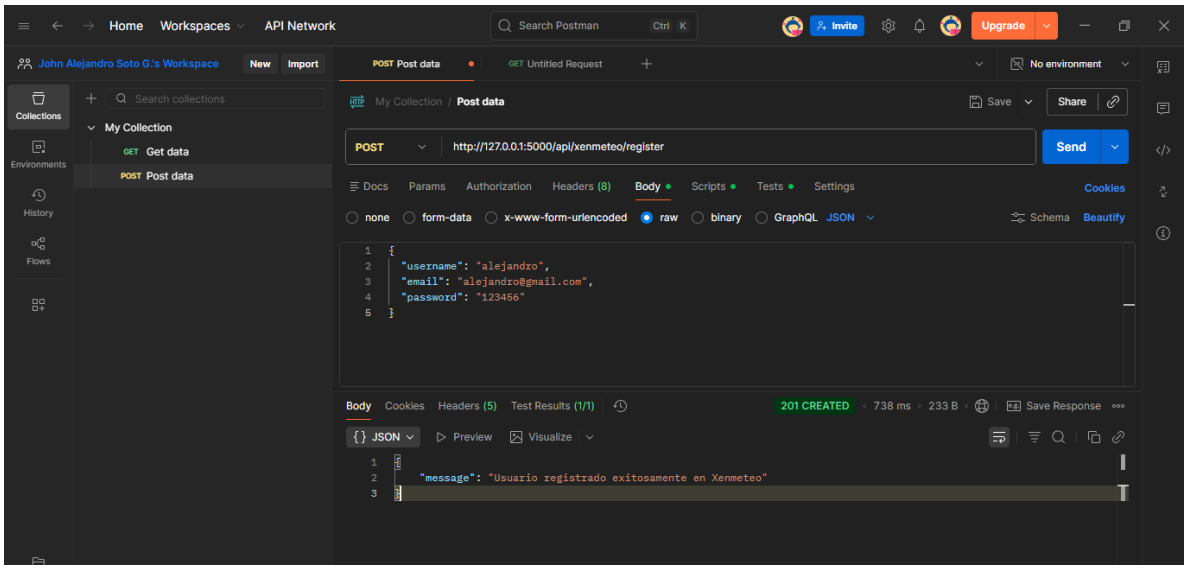
Ponemos la URL de conexión

http://127.0.0.1:5000/api/xenmeteo/register

Pegamos la autenticación JSON en Body



Luego de dar clic en Send > "message": "Usuario registrado exitosamente en Xenmeteo"



Creado en el código 201.

Ahora probamos Login

Cambiamos URL a http://127.0.0.1:5000/api/xenmeteo/login

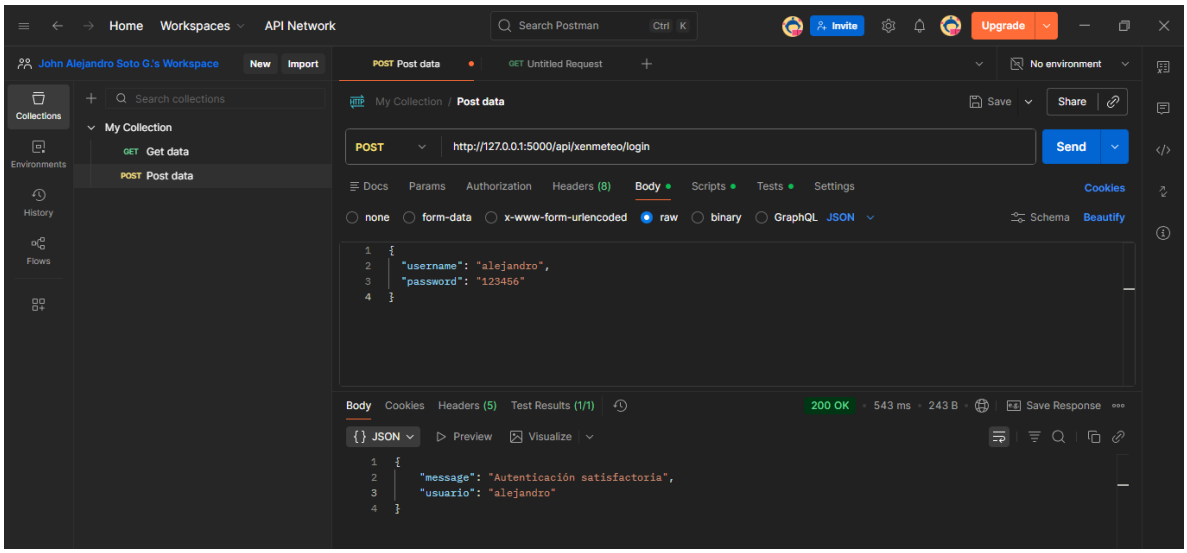
Y pegamos

{

"username": "alejandro",

"password": "123456"

}



Lo anterior con el fin de recibir respuestas mediante la herramienta Postman.

A continuación, iniciamos control de versiones en Git en Git Bash

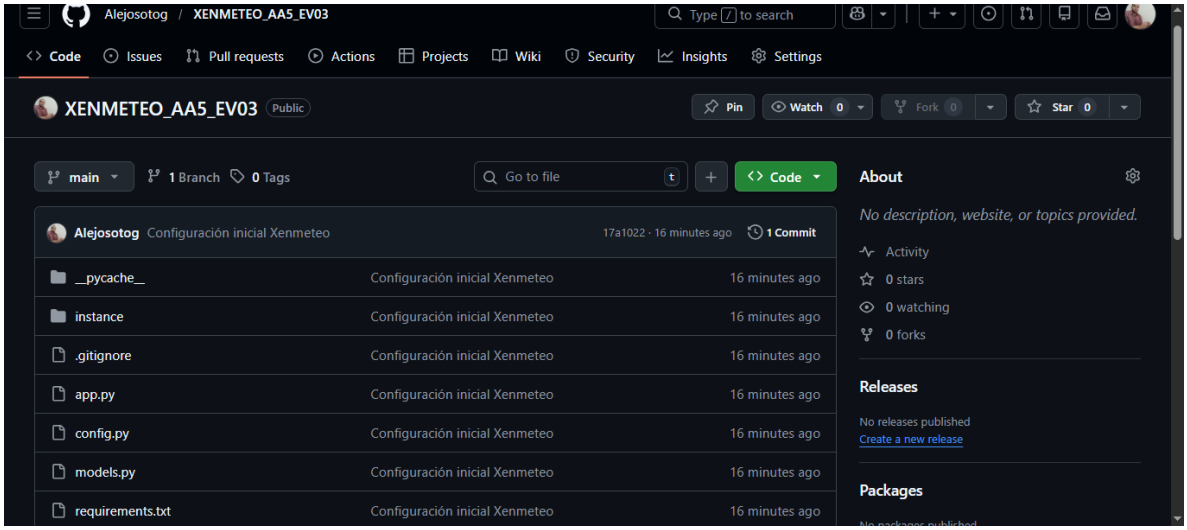
```
MINGW64:/c/Users/JOHN/Desktop/XENMETEO_AA5_EV03
JOHN@DESKTOP-MQPT3FQ MINGW64 ~ (master)
$ cd Desktop/XENMETEO_AA5_EV03

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ pwd
/c/Users/JOHN/Desktop/XENMETEO_AA5_EV03

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ git init
Reinitialized existing Git repository in C:/Users/JOHN/Desktop/XENMETEO_AA5_EV03/.git/

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ ^[[200~cd ~
bash: $'\E[200~cd': command not found

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ cd ~
rm -rf .git

JOHN@DESKTOP-MQPT3FQ MINGW64 ~
$ cd Desktop/XENMETEO_AA5_EV03

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ pwd
/c/Users/JOHN/Desktop/XENMETEO_AA5_EV03

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ git init
Reinitialized existing Git repository in C:/Users/JOHN/Desktop/XENMETEO_AA5_EV03/.git/

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        __pycache__/
        app.py
        config.py
        instance/
        models.py
        requirements.txt
```

```
JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ git add .

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (master)
$ git commit -m "Configuración inicial Xenmeteo"
[master (root-commit) 17a1022] Configuración inicial Xenmeteo
 8 files changed, 104 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 __pycache__/config.cpython-313.pyc
 create mode 100644 __pycache__/models.cpython-313.pyc
 create mode 100644 app.py
 create mode 100644 config.py
 create mode 100644 instance/xenmeteo.db
 create mode 100644 models.py
 create mode 100644 requirements.txt
```

Renombramos la rama actual a main y subimos la rama main al repositorio remoto

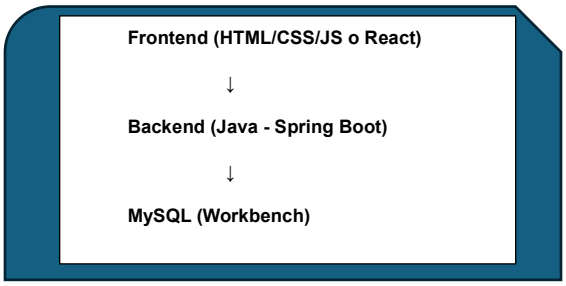
```
JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote -v
origin  https://github.com/Alejosotog/xenmeteo-aa5.git (fetch)
origin  https://github.com/Alejosotog/xenmeteo-aa5.git (push)

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote remove origin

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote -v

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote add origin https://github.com/Alejosotog/XENMETEO_AA5_EV03.git

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 4 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 3.52 KiB | 514.00 KiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Alejosotog/XENMETEO_AA5_EV03.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Viasulización del proyecto subido a Git-hub



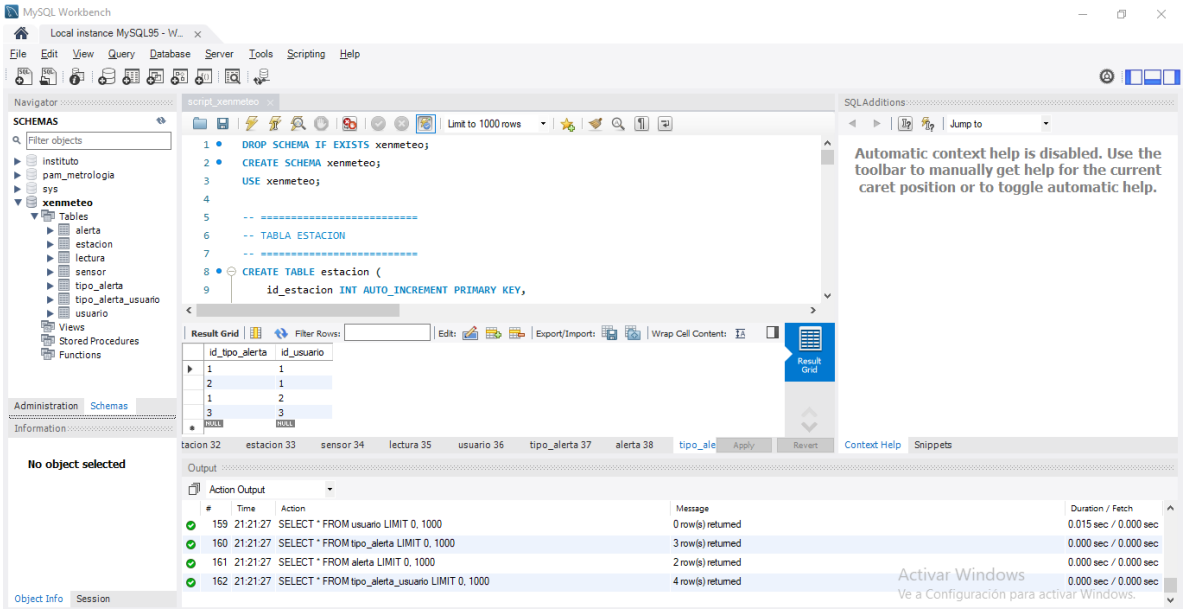
Ahora, pasaremos a hacer la conexión real después de este ejercicio donde se emuló el frontend por medio del postman

Objetivo



FASE 1 — Preparar la Base de Datos (MySQL Workbench)

Nos conectamos al servidor local con wokbench

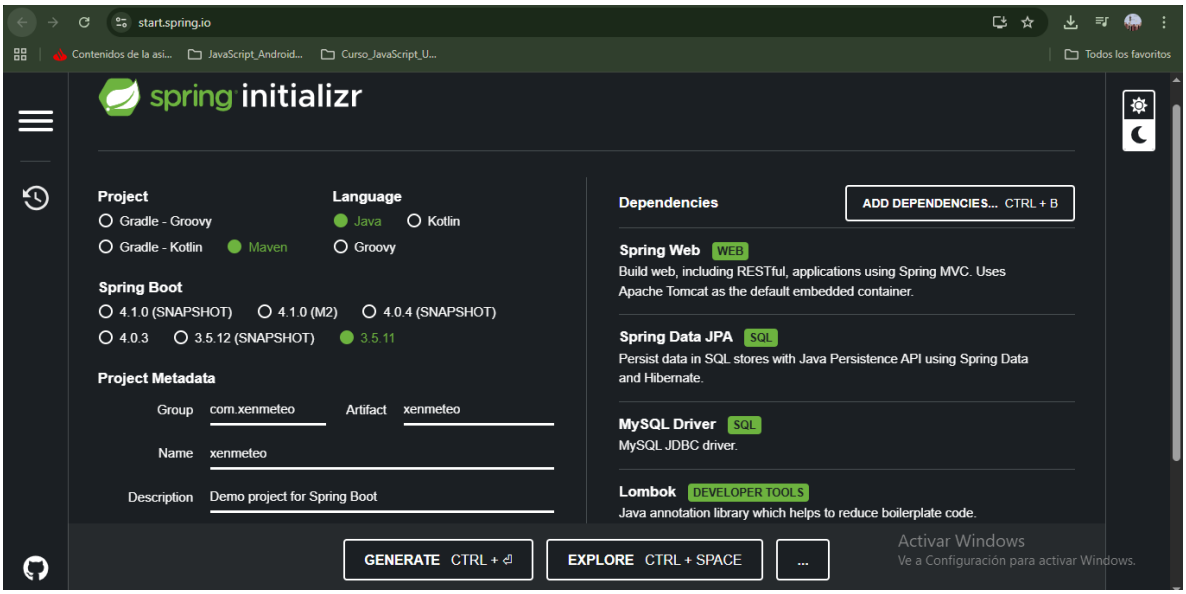


FASE 2 — Crea Backend con Spring Boot

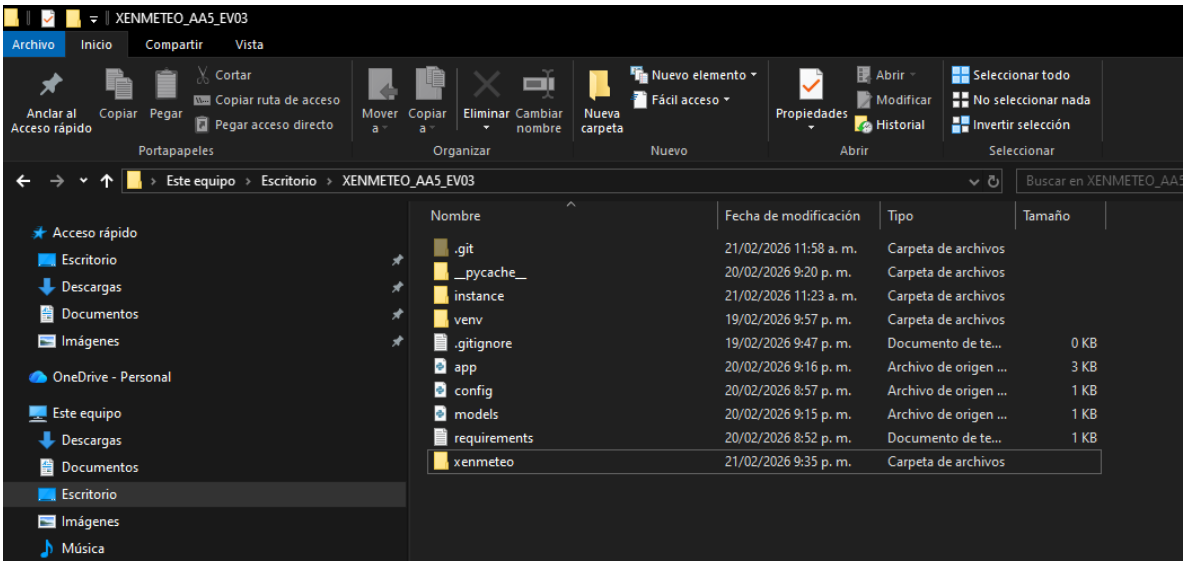
Creamos un proyecto en Spring Initializr

Configuramos EXACTAMENTE así:

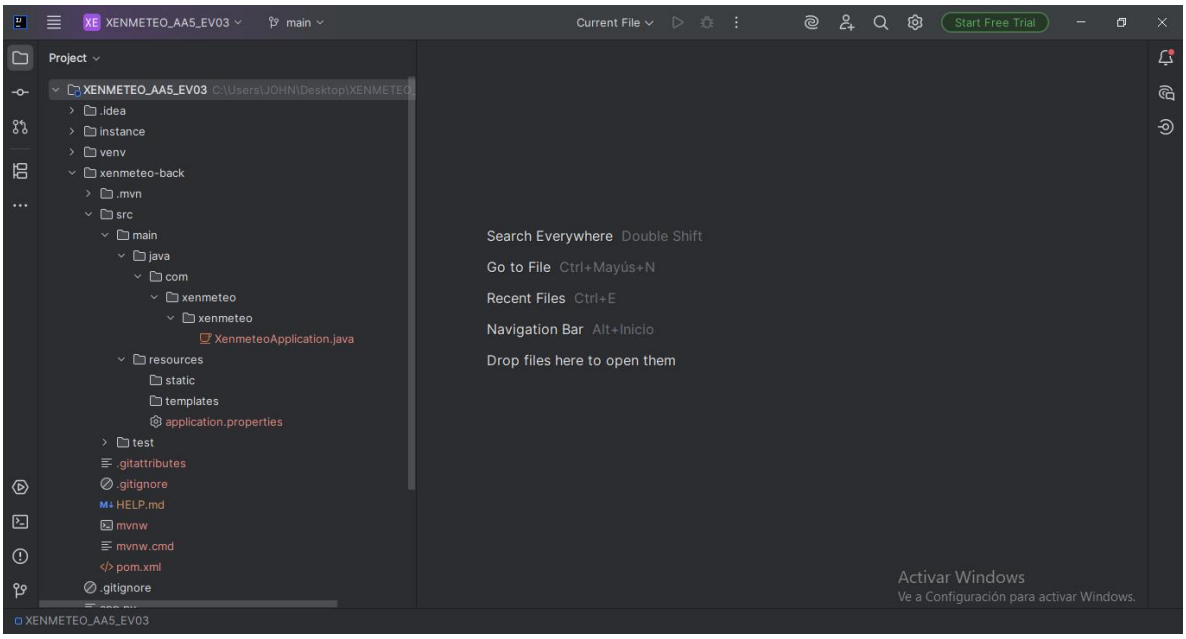
- Project: Maven
- Language: Java
- Spring Boot: última estable
- Group: com.xenmeteo
- Artifact: xenmeteo
- Packaging: Jar
- Java: 17



Al darle en generar ctrl + enter , se genera el backend que iría en el repositorio del proyecto



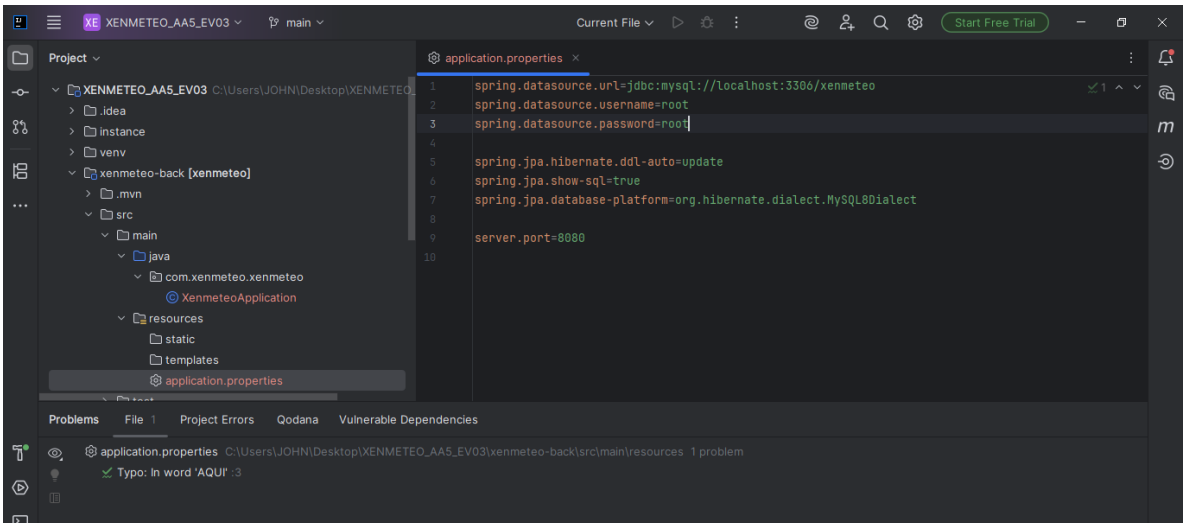
Abrimos el proyecto con IntelliJ IDEA

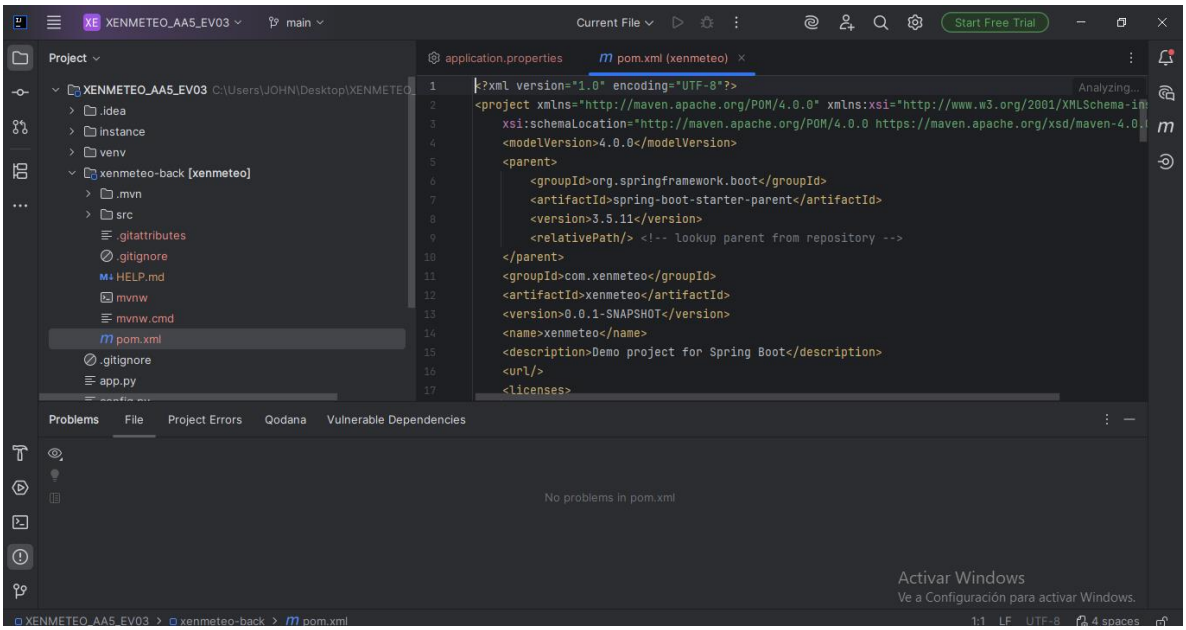


FASE 3 — Configurar conexión a MySQL

Abrir application.properties

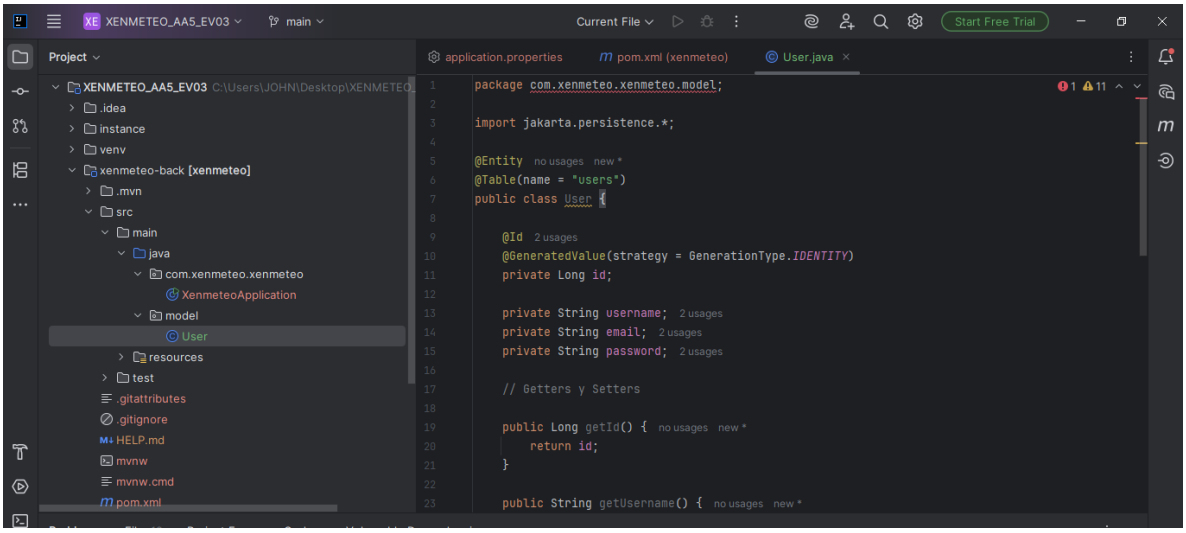
src/main/resources/application.properties





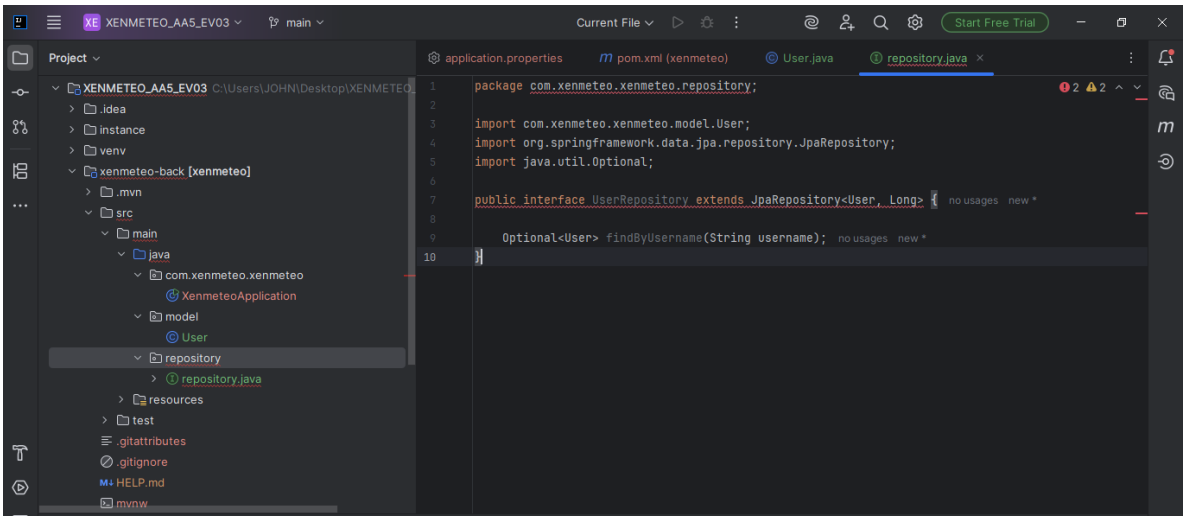
FASE 4 — Crear la Entidad User

Crear paquete model



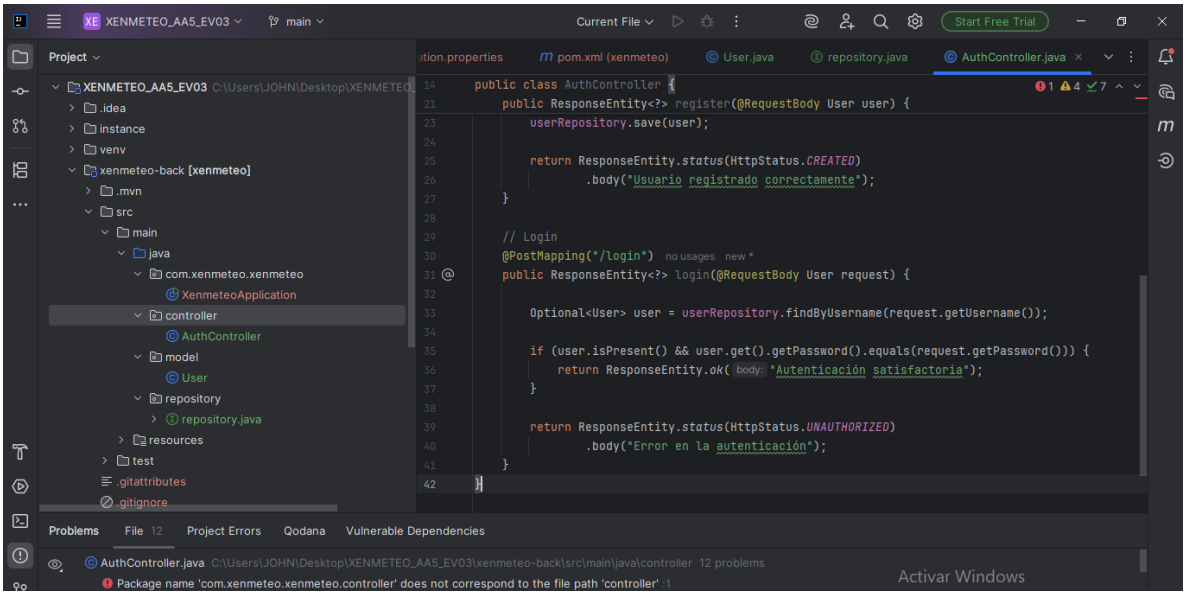
FASE 5 — Crear Repository

Crear paquete repository



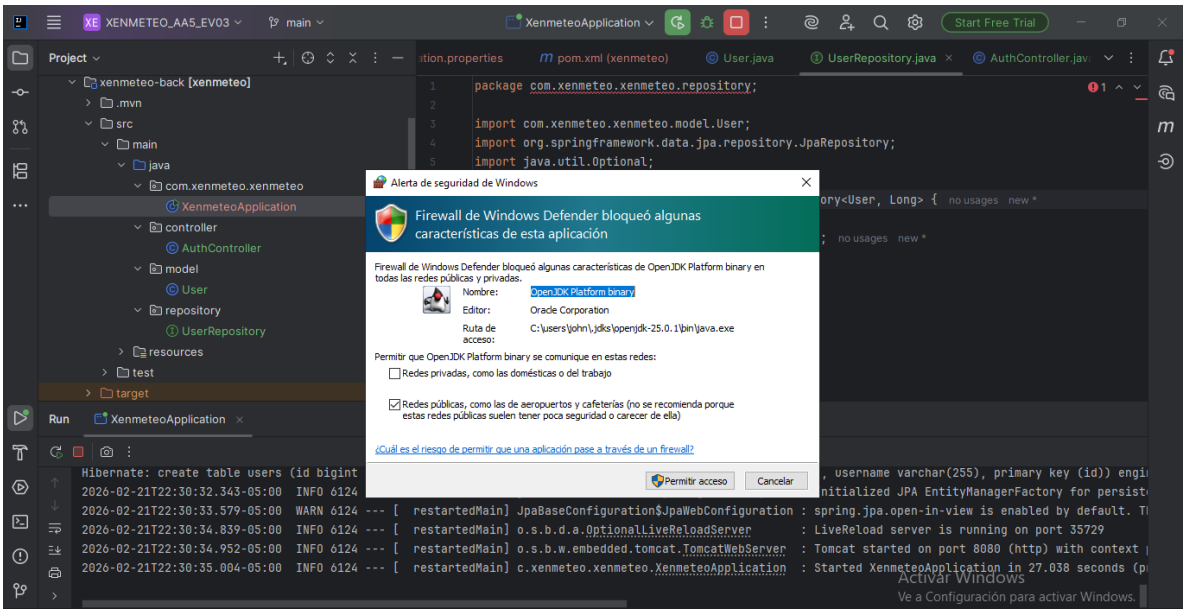
FASE 6 — Crear Controller

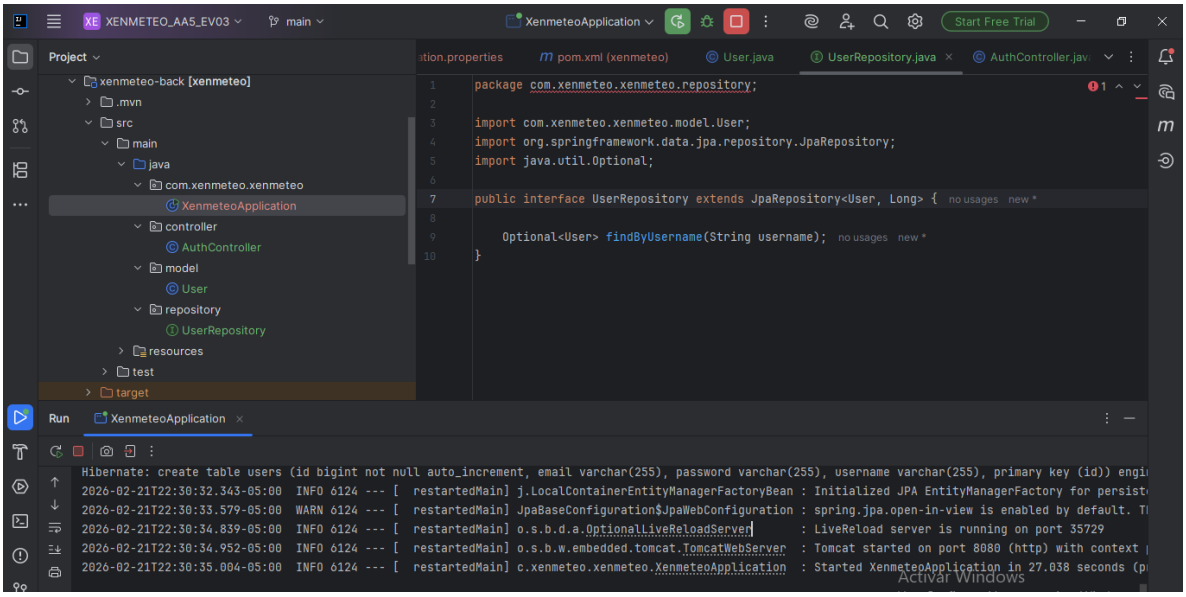
Crear paquete controller



FASE 7 — Ejecutar el Backend

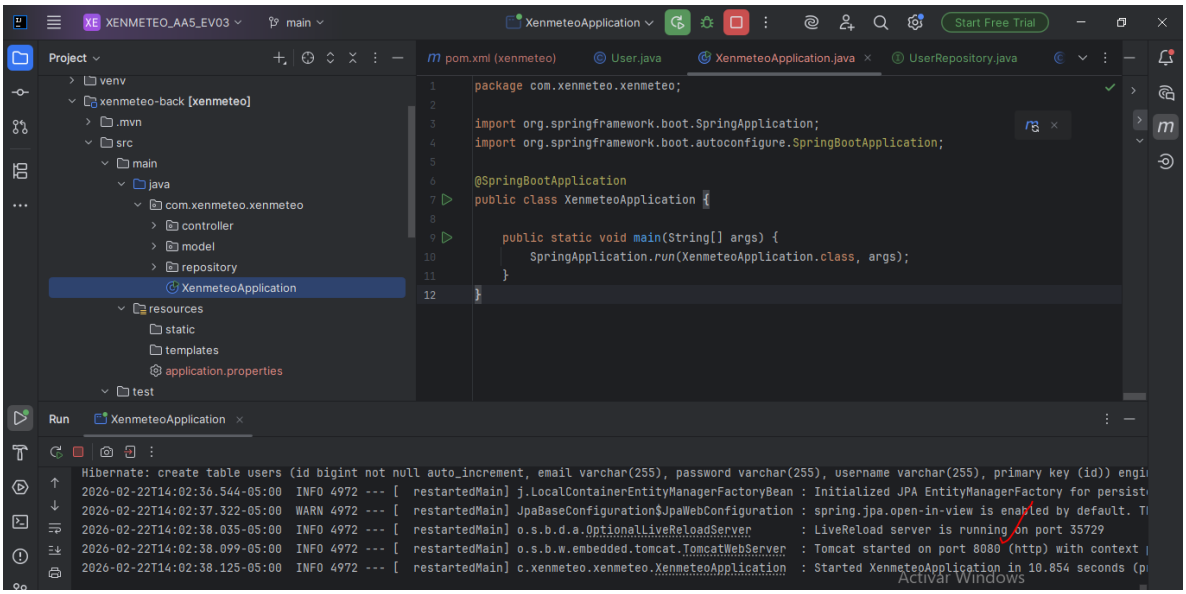
Clic derecho en XenmeteoAplicacion y luego en permitir acceso al OpenJDK



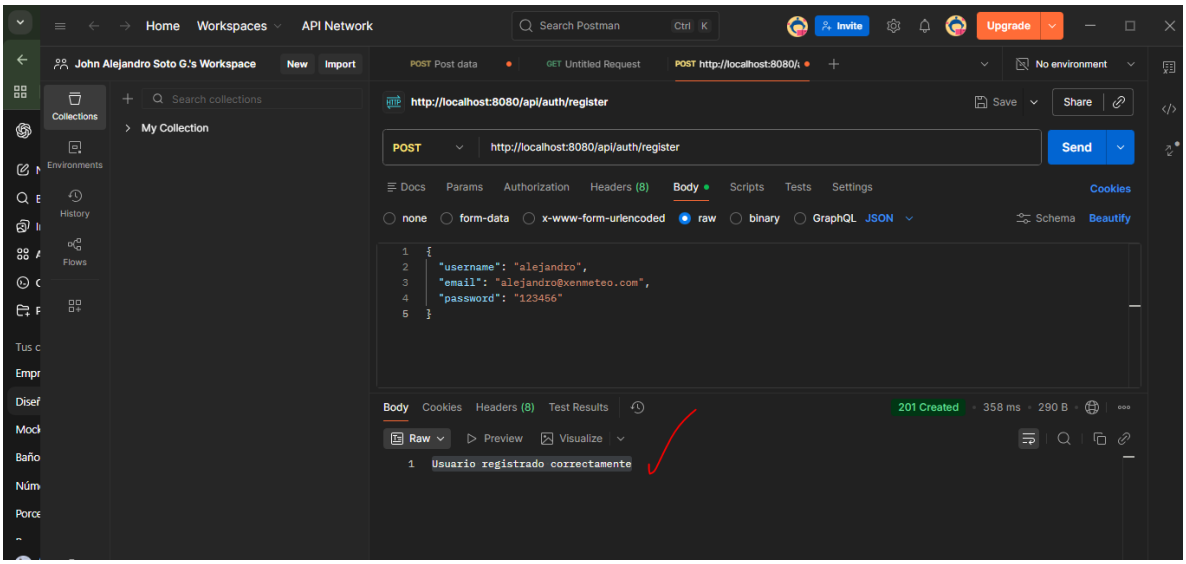


Con la anterior respuesta confirmamos:

El Spring Boot arrancó
El Tomcat está corriendo
Hibernate creó la tabla users
La conexión con MySQL funciona
El backend está activo



FASE 8 — Probar en Postman



La estructura del endpoint se forma de la siguiente manera:

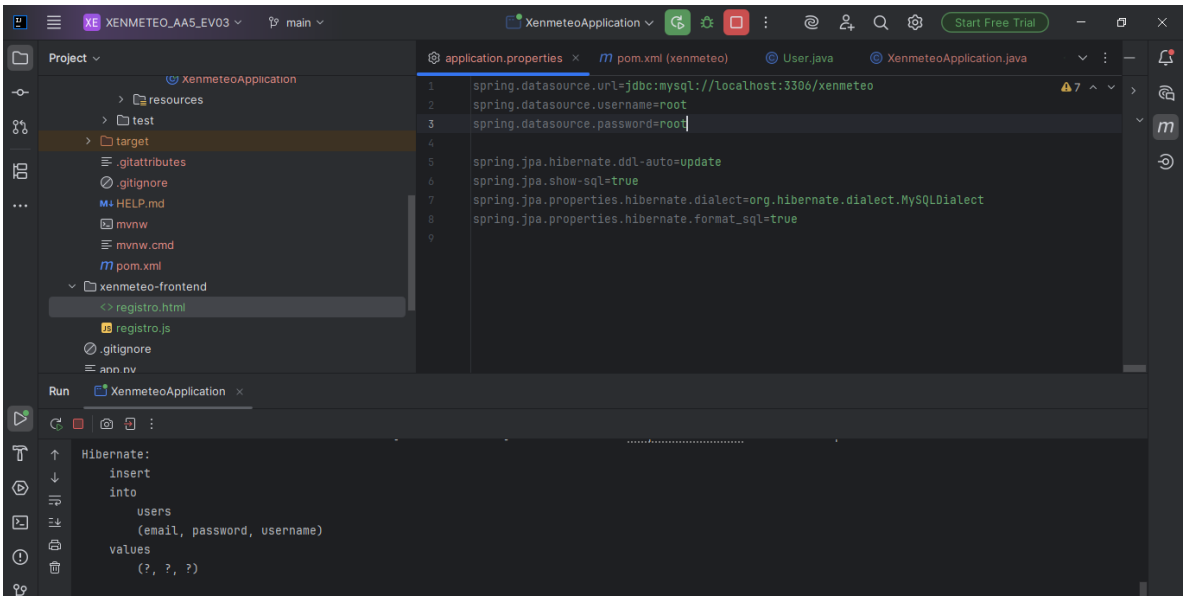
@RequestMapping("/api/auth") + @PostMapping("/register") = /api/auth/register

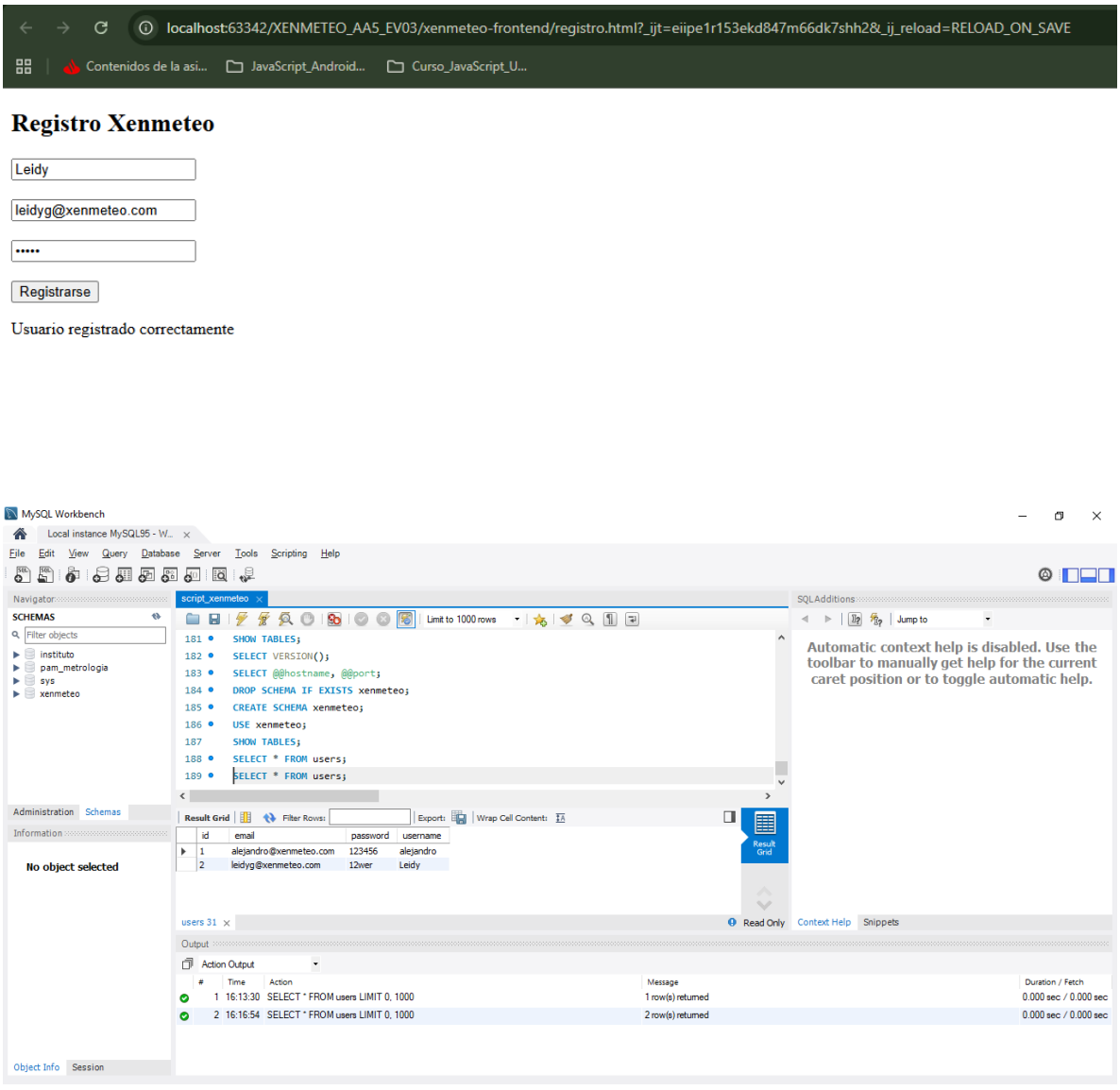
FASE 9 — Integrar con Frontend Xenmeteo

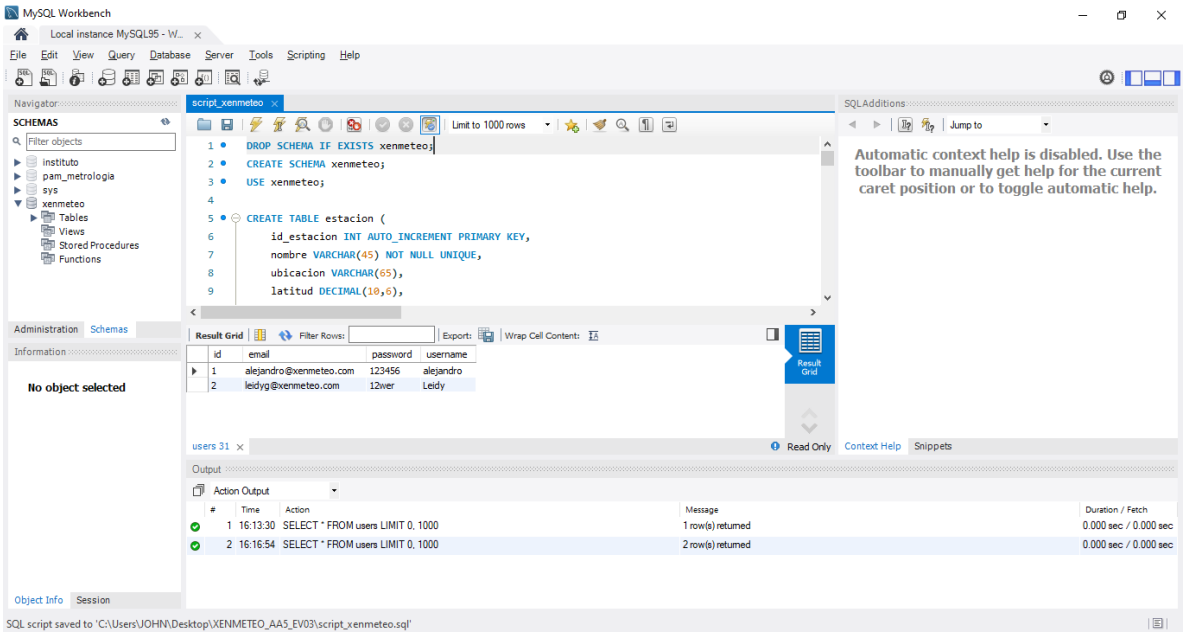
Arquitectura final

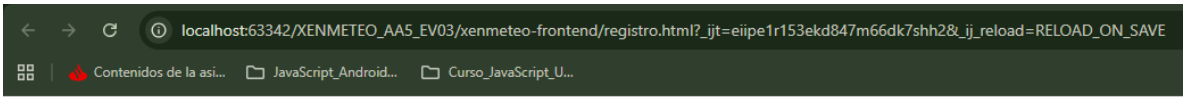
[Usuario] - [Página Xenmeteo (HTML + JS)] - (fetch) - [Backend Spring Boot] - [MySQL]

Creamos carpeta xenmeteo-frontend y dentro de estas creamos un archivo registro.hmtl y otro registro.js


XENMETEO_AA5_EV03
main
XenmeteoApplication
Start Free Trial
Project
XenmeteoApplication
resources
test
target
.gitattributes
.gitignore
HELP.md
mvnw
mvnw.cmd
pom.xml
xenmeteo-frontend
registro.html
registro.js
.gitignore
app.py
application.properties
pom.xml (xenmeteo)
User.java
XenmeteoApplication.java
spring.datasource.url=jdbc:mysql://localhost:3306/xenmeteo
spring.datasource.username=root
spring.datasource.password=root
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
spring.jpa.properties.hibernate.format_sql=true
Run
XenmeteoApplication
Hibernate:
insert
into
users
(email, password, username)
values
(?, ?, ?)


localhost:63342/XENMETEO_AA5_EV03/xenmeteo-frontend/registro.html?_ijt=eiipe1r153ekd847m66dk7shh2&_ij_reload=RELOAD_ON_SAVE
Contenidos de la asi...
JavaScript_Android...
Curso_JavaScript_U...
Registro Xenmeteo
Leidy
leidyg@xenmeteo.com
Registrarse
Usuario registrado correctamente
MySQL Workbench
Local instance MySQL95 - W...
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
instituto
pam_metrologia
sys
xenmeteo
Administration Schemas
Information
No object selected
Object Info Session
script_xenmeteo
Limit to 1000 rows
181 SHOW TABLES;
182 SELECT VERSION();
183 SELECT @@hostname, @@port;
184 DROP SCHEMA IF EXISTS xenmeteo;
185 CREATE SCHEMA xenmeteo;
186 USE xenmeteo;
187 SHOW TABLES;
188 SELECT * FROM users;
189 SELECT * FROM users;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
id email password username
1 alejandro@xenmeteo.com 123456 alejandro
2 leidyg@xenmeteo.com 12wer Leidy
users 31
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 16:13:30 SELECT * FROM users LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
2 16:16:54 SELECT * FROM users LIMIT 0, 1000 2 row(s) returned 0.000 sec / 0.000 sec





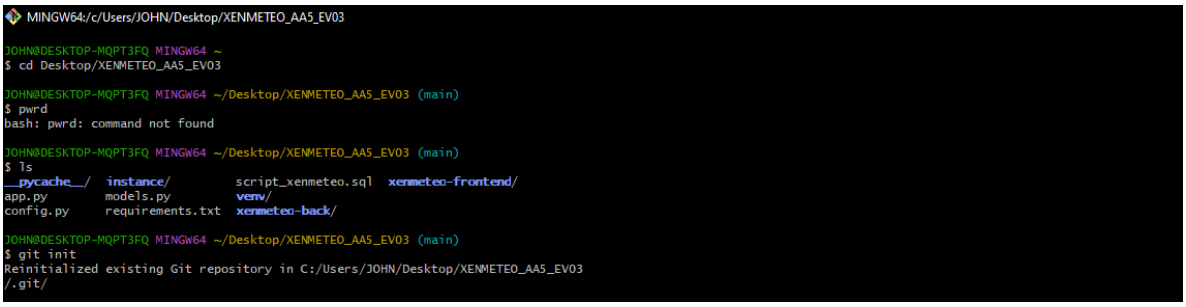
Registro Xenmeteo

Leidy

leidyg@xenmeteo.com

••••

Registrarse



```
JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/model/User
.java
        new file:   xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/repository
/UserRepository.java
        new file:   xenmeteo-back/src/main/java/xenmeteo/controller/AuthControll
er.java
        new file:   xenmeteo-frontend/registro.html
        new file:   xenmeteo-frontend/registro.js

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    xenmeteo-back/src/main/java/xenmeteo/controller/AuthControll
er.java
        modified:   xenmeteo-frontend/registro.html
        modified:   xenmeteo-frontend/registro.js

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .idea/
        script_xenmeteo.sql
        xenmeteo-back/.gitattributes
        xenmeteo-back/.gitignore
        xenmeteo-back/.mvn/
        xenmeteo-back/mvnw
        xenmeteo-back/mvnw.cmd
        xenmeteo-back/pom.xml
        xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/XenmeteoApplication.ja
va
        xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/controller/
        xenmeteo-back/src/main/resources/
        xenmeteo-back/src/test/
```

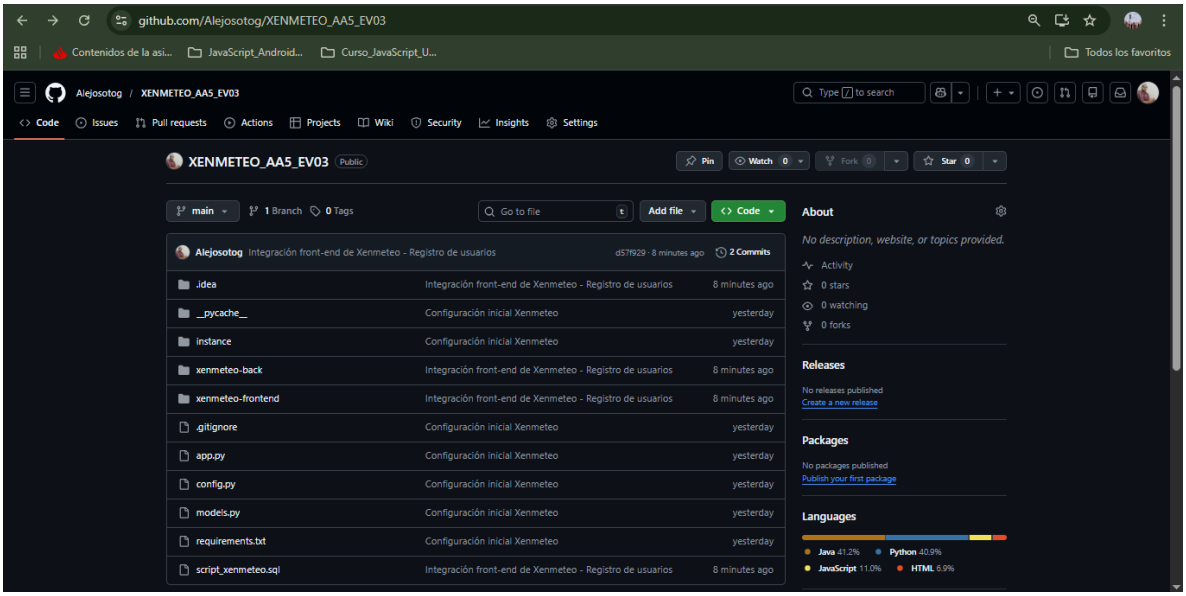
```
MINGW64:/c/Users/JOHN/Desktop/XENMETEO_AA5_EV03
warning: in the working copy of 'xenmeteo-back/.gitattributes', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/.mvn/wrapper/maven-wrapper.properties', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/mvnw.cmd', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/XenmeteoApplication.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/src/main/resources/application.properties', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'xenmeteo-back/src/test/java/com/xenmeteo/xenmeteo/XenmeteoApplicationTests.java', LF will be replaced by CRLF the next time Git touches it

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git commit -m "Integración front-end de Xenmeteo - Registro de usuarios"
[main d57f929] Integración front-end de Xenmeteo - Registro de usuarios
 23 files changed, 1066 insertions(+)
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/XENMETEO_AA5_EV03.iml
 create mode 100644 .idea/compiler.xml
 create mode 100644 .idea/encodings.xml
 create mode 100644 .idea/jarRepositories.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 .idea/modules.xml
 create mode 100644 .idea/vcs.xml
 create mode 100644 script_xenmeteo.sql
 create mode 100644 xenmeteo-back/.gitattributes
 create mode 100644 xenmeteo-back/.gitignore
 create mode 100644 xenmeteo-back/.mvn/wrapper/maven-wrapper.properties
 create mode 100644 xenmeteo-back/mvnw
 create mode 100644 xenmeteo-back/mvnw.cmd
 create mode 100644 xenmeteo-back/pom.xml
 create mode 100644 xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/XenmeteoApplication.java
 create mode 100644 xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/controller/AuthController.java
 create mode 100644 xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/model/User.java
 create mode 100644 xenmeteo-back/src/main/java/com/xenmeteo/xenmeteo/repository/UserRepository.java
 create mode 100644 xenmeteo-back/src/main/resources/application.properties
 create mode 100644 xenmeteo-back/src/test/java/com/xenmeteo/xenmeteo/XenmeteoApplicationTests.java
 create mode 100644 xenmeteo-frontend/registro.html
 create mode 100644 xenmeteo-frontend/registro.js

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git add .

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git commit -m "Integración front-end de Xenmeteo - Registro de usuarios"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

```
JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote remote origin
error: unknown subcommand: `remote'
usage: git remote [-v | --verbose]
   or: git remote add [-t <branch>] [-m <master>] [-f] [--tags | --no-tags] [--mirror=<fetch|push>] <name> <url>
   or: git remote rename [--[no-]progress] <old> <new>
   or: git remote remove <name>
   or: git remote set-head <name> (-a | --auto | -d | --delete | <branch>)
   or: git remote [-v | --verbose] show [-n] <name>
   or: git remote prune [-n | --dry-run] <name>
   or: git remote [-v | --verbose] update [-p | --prune] [(<group> | <remote>)...]
   or: git remote set-branches [--add] <name> <branch>...
   or: git remote get-url [--push] [--all] <name>
   or: git remote set-url [--push] <name> <newurl> [<oldurl>]
   or: git remote set-url --add <name> <newurl>
   or: git remote set-url --delete <name> <url>

    -v, --[no-]verbose    be verbose; must be placed before a subcommand


JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote remove origin

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote -v

JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git remote add origin https://github.com/Alejosotog/XENMETEO_AA5_EV03.git
```

Finalmente, subimos los cambios al repositorio remoto de Alejosotog – Xenmeteo_AA5_EV03

```
JOHN@DESKTOP-MQPT3FQ MINGW64 ~/Desktop/XENMETEO_AA5_EV03 (main)
$ git push -u origin main
Enumerating objects: 46, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 4 threads
Compressing objects: 100% (33/33), done.
Writing objects: 100% (45/45), 15.00 KiB | 1.36 MiB/s, done.
Total 45 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Alejosotog/XENMETEO_AA5_EV03.git
   17a1022..d57f929  main -> main
branch 'main' set up to track 'origin/main'.
```

https://github.com/Alejosotog/XENMETEO_AA5_EV03



Conclusiones

La integración entre frontend, backend y base de datos permitió validar el funcionamiento completo del sistema Xenmeteo en su módulo de registro de usuarios. Se comprobó que la comunicación entre capas es efectiva, que el backend procesa correctamente las solicitudes HTTP y que la persistencia de datos en MySQL se realiza de manera adecuada mediante JPA e Hibernate.

Este ejercicio fortalece la comprensión de la arquitectura en capas, la conexión a bases de datos relacionales y el desarrollo de APIs RESTful. Además, sienta las bases para futuras mejoras como autenticación segura, validaciones avanzadas y despliegue en entornos productivos.

Referencias

[1] Spring Boot. (2024). Spring Boot Reference Documentation. Disponible en: https://spring.io/projects/spring-boot

[2] MySQL. (2024). MySQL 8.0 Reference Manual. Oracle Corporation. Disponible en: https://dev.mysql.com/doc/

[3] Hibernate. (2024). Hibernate ORM Documentation. Disponible en: https://hibernate.org/orm/documentation/